Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
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Abstract

1. Problem. Student treasurers must complete an accountability report (LPJ) before faculty move subsequent proposals funding workshops and peer learning. Rules differ by campus: original stamps, attendance sheets per meal, matching totals, revenue stamps above Rp5,000,000 (Law 10/2020 Art. 3), withholding, and strict deadlines. At FILKOM UB, incomplete submissions delay subsequent events. Treasurers lack accounting training, so study hours go into rework. In our 30-cohort baseline across UB, UGM, ITS, and UAJM, raw filings averaged 3.7 mechanical defects, causing 100% initial desk-review rejections.

2. Intended Users. Primary users are treasurers and secretaries of program associations, BEM, and clubs. The faculty reviewer reads the file. RapiLPJ does not release funds.

3. Supporting Observations or Evidence. FIB UB requires stamped originals, one attendance sheet per meal session, and matching totals. FEB UGM requires a Rp10,000 stamp above Rp5,000,000 and a scan when thermal paper fades. An ITS department sets ten days; ITS Biology requires one line per receipt with tax included. A 2021 UAJM study describes unstandardized activity documentation. FILKOM UB's 2026 circular records repeated withholding and completeness failures.

4. Proposed Solution and Core User Flow. RapiLPJ is a practice run: rejection happens on a laptop, not at the faculty desk. Gemini Flash on Vertex AI (Singapore) reads vendor, date, invoice number, nominal, and stamp presence into JSON and does no arithmetic. The treasurer confirms each field beside the slip. Every change stays in an edit log. Offline fixtures need no API key. A client-side kernel flags duplicate invoices, dates outside the activity window, a missing or wrong stamp above Rp5,000,000, and variance above 10 percent. Meal caps and PPh 21 come only from a rule card to be signed by two advisors. A missing rate returns preset gap and names what to ask the faculty, instead of a guessed tax that would be rejected. Export stays locked until one question matches the statute — exactly Rp5,000,000 is not dutiable; above it is — and any variance is written on a fixed template. A SHA-256 covers the confirmed ledger, edit log, quiz, addendum, and finding ids. The verifier recomputes it. The hash only shows the file was not changed after the checks.

5. How It Differs from Existing Alternatives. A raw LLM mis-adds. A spreadsheet can list the same rules and still export a wrong stamp, a guessed tax, or a silent edit. Campus portals collect files after submission and miss the defect before the next proposal is frozen.

6. Planned Technologies. Static bilingual HTML, CSS, and JavaScript on Firebase Hosting. No application database. Open JavaScript kernel, FileReader, a .lpj JSON file, and Web Crypto SHA-256. WCAG 2.1 AA.

7. Expected Benefit and Development Plan. The semifinal is one activity: perception, confirmation, rule card, quiz, and desk-check. Final Day adds several slips and a QR re-check. Across 30 simulated cohorts, the kernel resolved all 112 mechanical defects (13 missing stamps, 15 thermal slips, 20 duplicate invoices), cutting verification to 15.6 ms per dossier with 100 percent verifier clearance. The main risk is a misread thermal slip. Confirmation and the edit log are the control. Arithmetic never trusts the model.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026.